

Faculty of Computing



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Sap: 64281

[Computer Communications & Network]

Lab Manual

LAB - 1

Course	Computer Communications and Network Lab
Course Instructor	Ms. Saleha Nasim
Credit Hour	1
Lab No.	1
Lab Subject	Computer Network Configuration Commands
Lab Objective	1. Introduction to IP addressing 2. Identify tools used for discovering a computer's network configuration with various operating systems. 3. Gather information, including the connection, host name, MAC (Layer 2) Address, and TCP/IP Network (layer 3) 4. Compare the network information to that of other PCs on the network. 5. Learn to use the TCP/IP packet Internet groper (ping) command from a workstation. 6. Learn to use the trace route (tracert) command from a workstation. 7. Observe name-resolution occurrences using WINS and DNS servers.
Tools	PC/Laptop, Lab Manual

Background

This lab assumes that you are using Windows 11 or Windows 10. This is nondestructive lab that you can perform on any host without changing the system's configuration.

Ideally, you perform this lab in a LAN environment that connects to the Internet. You can use a single remote connection via a dialup modem or DSL. You will need the IP address information which the instructor should provide.

IP Addressing

IP address stands for "Internet Protocol address." The Internet Protocol is a set of rules for communication over the internet, such as sending mail, streaming video, or connecting to a

website. An IP address, or Internet Protocol address, is a series of numbers that identifies any device on a network. Computers use IP addresses to communicate with each other both over the internet as well as on other networks.

IP addresses are the identifier that allows information to be sent between devices on a network: they contain location information and make devices accessible for communication. The internet needs a way to differentiate between different computers, routers, and websites. IP addresses provide a way of doing so and form an essential part of how the internet works.

An IP address is a string of numbers separated by periods. IP addresses are expressed as a set of four numbers — an example address might be 192.158.1.38. Each number in the set can range from 0 to 255. So, the full IP addressing range goes from 0.0.0.0 to 255.255.255.255.

The internet protocols manage the process of assigning each unique device its own IP address. (Internet protocols do other things as well, such as routing internet traffic.) This way, it's easy to see which devices on the internet are sending, requesting, and receiving what information.

IP addresses are like telephone numbers, and they serve the same purpose. When you contact someone, your phone number identifies who you are, and it assures the person who answers the phone that you are who you say you are. IP addresses do the exact same thing when you're online — that's why **every single device that is connected to the internet has an IP address**.

There are two types of IP addresses: IPv4 and IPv6. It's easy to recognize the difference if you count the numbers. IPv4 addresses contain a series of four numbers, ranging from 0 (except the first one) to 255, each separated from the next by a period — such as 5.62.42.77.

IPv6 addresses are represented as eight groups of four hexadecimal digits, with the groups separated by colons. A typical IPv6 address might look like this: 2620:0aba2:0d01:2042:0100:8c4d:d370:72b4.

Network Configuration

Part A Using ipconfig command

Step 1: Connect to the Internet

Establish and verify connectivity to the Internet. This step ensures the computer has an IP address.

Step 2: Gather TCP/IP configuration information

- a. Windows NT/2000/XP: Use the start menu to open the command prompt (Start/Programs/Accessories/Command Prompt or Start/Programs/Command Prompt)

Windows 10: Press Window+R to open Run box. Then click ok to run a regular command prompt.

- b. Type ipconfig and press Enter key.

```
Microsoft Windows [Version 10.0.22631.5472]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ken tech>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv4 Address. . . . . : 10.55.97.109
    Subnet Mask . . . . . : 255.255.255.0
```

- c. The screen shows the ip address, subnet mask and the default gateway. The ip address and the default gateway should be in the same network or subnet. Otherwise the host wouldn't be able to communicate outside the network.

Step 3: Record the following TCP/IP information for this computer

- a. IP address: 192.168.56.1
- b. Subnet mask: 255.255.255.0
- c. Default Gateway: 10.55.97.231(this is from ipconfig/all)

Step 4: Compare this computer's TCP/IP configuration to that of others on the LAN

If this computer is on LAN, compare the information of several machines (Hosts).

a. Are there any similarities?

Yes, all computers on the LAN share similar subnet masks, use the same default gateway, and have IP addresses in the same network range.

b. What is similar about the IP address?

The first three octets (e.g., 192.168.1.x) are usually the same, indicating they are on the same local network.

c. What is similar about the default gateway?

All computers use the same default gateway, typically the IP address of the router (e.g., 192.168.1.1).

d. Is there anything similar in subnet mask?

Yes, all devices typically use the same subnet mask (e.g., 255.255.255.0), defining the network size.

e. Record a couple of the IP addresses (of your nearby hosts)

```
C:\Users\ken tech>arp -a

Interface: 192.168.56.1 --- 0xa
    Internet Address      Physical Address         Type
    192.168.56.255        ff-ff-ff-ff-ff-ff       static
    224.0.0.22            01-00-5e-00-00-16       static
    224.0.0.251           01-00-5e-00-00-fb       static
    224.0.0.252           01-00-5e-00-00-fc       static
    239.255.255.250       01-00-5e-7f-ff-fa       static

Interface: 10.55.97.109 --- 0xf
    Internet Address      Physical Address         Type
    10.55.97.231          d2-34-9c-bd-56-92       dynamic
    10.55.97.255          ff-ff-ff-ff-ff-ff       static
    224.0.0.22            01-00-5e-00-00-16       static
    224.0.0.251           01-00-5e-00-00-fb       static
    224.0.0.252           01-00-5e-00-00-fc       static
    239.255.255.250       01-00-5e-7f-ff-fa       static
    255.255.255.255       ff-ff-ff-ff-ff-ff       static

C:\Users\ken tech>
```

Step 5: Check Additional TCP/IP configuration information

- a. To see more information, type ipconfig/all and press Enter key. The next figure shows the detailed IP configuration of this computer on the screen.**

```
C:\Users\ken tech>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-GS7MTG1
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) Ethernet Connection (6) I219-LM
    Physical Address. . . . . : 38-22-E2-4A-5C-6E
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix . :
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter
    Physical Address. . . . . : 0A-00-27-00-00-0A
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.56.1(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
    NetBIOS over Tcpip. . . . . : Enabled

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : TAP-Windows Adapter V9
    Physical Address. . . . . : 00-FF-AD-42-B8-B5
```

```
Physical Address. . . . . : 00-FF-AD-42-B8-B5
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 60-F2-62-15-1C-89
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 62-F2-62-15-1C-88
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
Physical Address. . . . . : 60-F2-62-15-1C-88
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 10.55.97.109(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, August 21, 2025 7:42:59 PM
Lease Expires . . . . . : Thursday, August 21, 2025 8:42:58 PM
Default Gateway . . . . . : 10.55.97.231
DHCP Server . . . . . : 10.55.97.231
DNS Servers . . . . . : 10.55.97.231
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 60-F2-62-15-1C-8C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\ken tech>
```

b. You should see the following information:

- a. The host name
- b. The physical address of this machine
- c. IP address

- d. Subnet mask
 - e. Default gateway
 - f. DNS servers
- c. In the LAN, compare your results with a few nearby computers. What similarities do you see in the physical (MAC) address?

- Use `ipconfig /all` on several nearby computers to check their **MAC addresses** (also called **Physical Addresses**).
- The MAC address is listed like this: Physical Address. : 00-14-22-01-23-45

While MAC addresses are **unique** to each device, they may share **some similarities**, especially if the devices are from the **same manufacturer** (e.g., all from Dell, HP, etc.). The **first half** of a MAC address often indicates the **vendor**.

- d. Write down the computer's host name: DESKTOP-GS7MTG1

-
- e. Write down the host name of couple of other computers.

On each nearby computer, open **Command Prompt** and type: `hostname`

Example Host Names:

- I. DESKTOP-01
- II. OFFICE-PC
- III. STUDENT-LAPTOP
- IV. RIPHAH

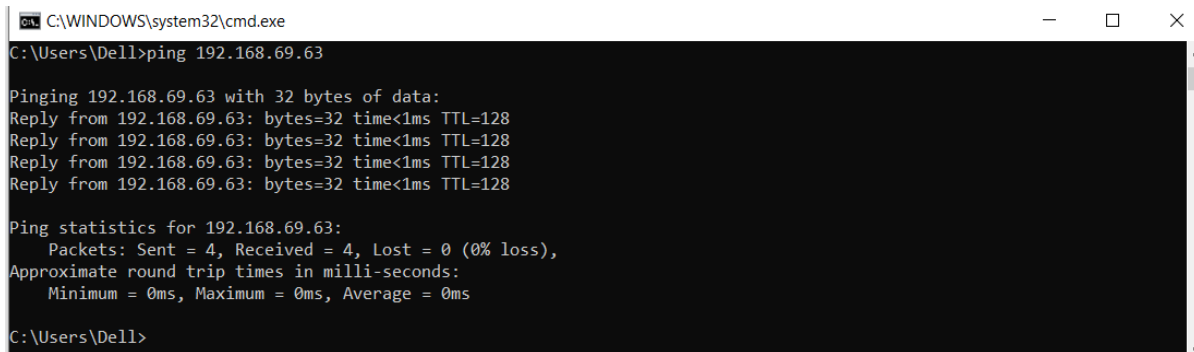
Step 6: Close the screen when finished.

Part B Using PING TRACERT from a workstation

Establish and verify connectivity to the internet. This step ensures that the computer has an IP address.

Step 1:

- a. Open the command prompt.
- b. Type ping, give space and type the IP address of a computer recorded in the previous lab.



```
C:\WINDOWS\system32\cmd.exe
C:\Users\Dell>ping 192.168.69.63

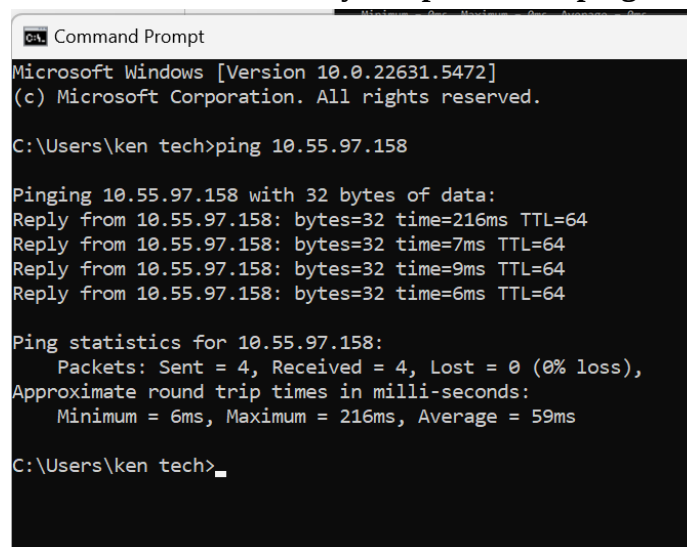
Pinging 192.168.69.63 with 32 bytes of data:
Reply from 192.168.69.63: bytes=32 time<1ms TTL=128
Reply from 192.168.69.63: bytes=32 time<1ms TTL=128
Reply from 192.168.69.63: bytes=32 time<1ms TTL=128
Reply from 192.168.69.63: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.69.63:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Dell>
```

Ping uses the Internet Control Message Protocol (ICMP) echo-request and echo-reply feature to test physical connectivity. Because ping report on four attempts, it gives an indication of the reliability of the connection. Look over the result and verify that the ping was successful. Was the ping successful? If not, report to the instructor.

- c. Ask the IP address of the nearby computers and ping. Note the result.



```
Command Prompt
Microsoft Windows [Version 10.0.22631.5472]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ken tech>ping 10.55.97.158

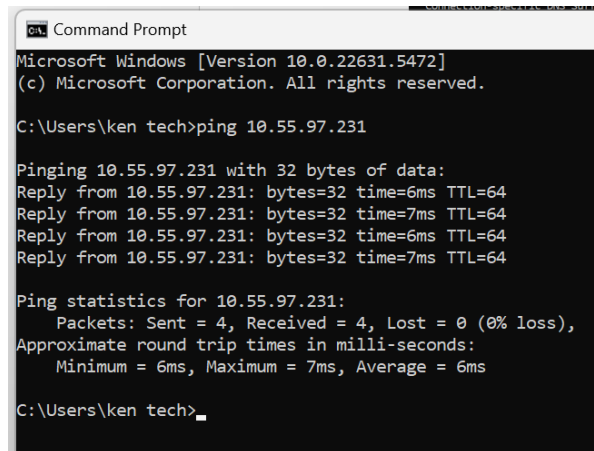
Pinging 10.55.97.158 with 32 bytes of data:
Reply from 10.55.97.158: bytes=32 time=216ms TTL=64
Reply from 10.55.97.158: bytes=32 time=7ms TTL=64
Reply from 10.55.97.158: bytes=32 time=9ms TTL=64
Reply from 10.55.97.158: bytes=32 time=6ms TTL=64

Ping statistics for 10.55.97.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 216ms, Average = 59ms

C:\Users\ken tech>
```

- d. Ping the IP address of default gateway and DNS server. Were the results successful?

Dns & Default gateway:



```
Command Prompt
Microsoft Windows [Version 10.0.22631.5472]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ken tech>ping 10.55.97.231

Pinging 10.55.97.231 with 32 bytes of data:
Reply from 10.55.97.231: bytes=32 time=6ms TTL=64
Reply from 10.55.97.231: bytes=32 time=7ms TTL=64
Reply from 10.55.97.231: bytes=32 time=6ms TTL=64
Reply from 10.55.97.231: bytes=32 time=7ms TTL=64

Ping statistics for 10.55.97.231:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 7ms, Average = 6ms

C:\Users\ken tech>
```

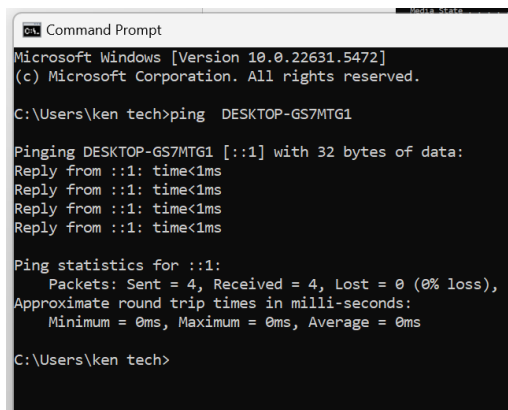
- e. Ping the Loop back ip address. Type the following address.

Ping 127.0.0.1

If loop back testing is successful then the TCP/IP is properly installed and functioning on this computer.

Is it successful? ___YES_____

- f. Ping the host name of the computer that you recorded in previous part.



```
Command Prompt
Microsoft Windows [Version 10.0.22631.5472]
(c) Microsoft Corporation. All rights reserved.

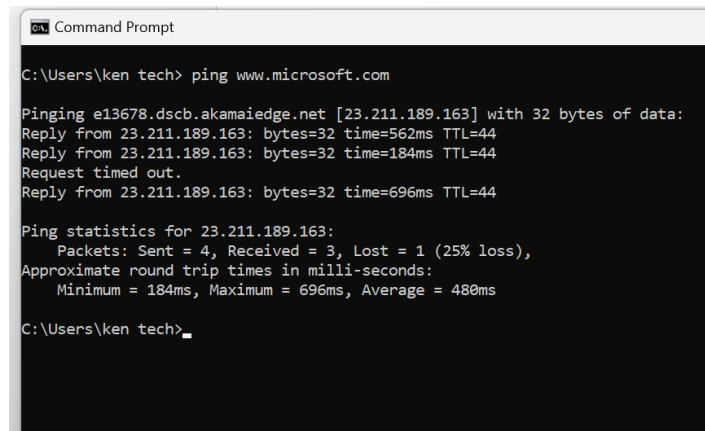
C:\Users\ken tech>ping DESKTOP-GS7MTG1

Pinging DESKTOP-GS7MTG1 [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\ken tech>
```

g. Ping the Microsoft website www.microsoft.com



```
Command Prompt
C:\Users\ken tech> ping www.microsoft.com

Pinging e13678.dscb.akamaiedge.net [23.211.189.163] with 32 bytes of data:
Reply from 23.211.189.163: bytes=32 time=562ms TTL=44
Reply from 23.211.189.163: bytes=32 time=184ms TTL=44
Request timed out.
Reply from 23.211.189.163: bytes=32 time=696ms TTL=44

Ping statistics for 23.211.189.163:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 184ms, Maximum = 696ms, Average = 480ms

C:\Users\ken tech>
```

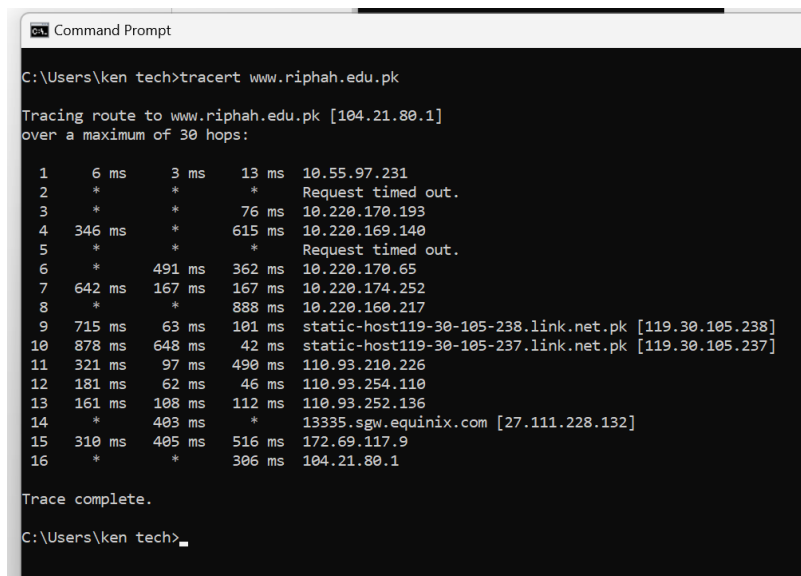
Step 3: Trace the route to Riphah International University's Website

Type `tracert www.riphah.edu.pk` and press Enter key

The result shows the complete route to the website and the number of hops in the path.

Trace a local host name or IP address in your local area network (LAN).

Record the output and interpret.



```
Command Prompt
C:\Users\ken tech>tracert www.riphah.edu.pk

Tracing route to www.riphah.edu.pk [104.21.80.1]
over a maximum of 30 hops:

  0  6 ms   3 ms   13 ms  10.55.97.231
  1  *      *      *      Request timed out.
  2  *      *      *      Request timed out.
  3  *      *      *      Request timed out.
  4  346 ms  *      615 ms  10.220.170.193
  5  *      *      *      Request timed out.
  6  *      491 ms 362 ms  10.220.170.65
  7  642 ms 167 ms 167 ms  10.220.174.252
  8  *      *      888 ms  10.220.160.217
  9  715 ms 63 ms 101 ms  static-host119-30-105-238.link.net.pk [119.30.105.238]
 10 878 ms 648 ms 42 ms  static-host119-30-105-237.link.net.pk [119.30.105.237]
 11 321 ms 97 ms 490 ms  110.93.210.226
 12 181 ms 62 ms 46 ms  110.93.254.110
 13 161 ms 108 ms 112 ms  110.93.252.136
 14 *      403 ms *      13335.sgw.equinix.com [27.111.228.132]
 15 310 ms 405 ms 516 ms  172.69.117.9
 16 *      *      306 ms  104.21.80.1

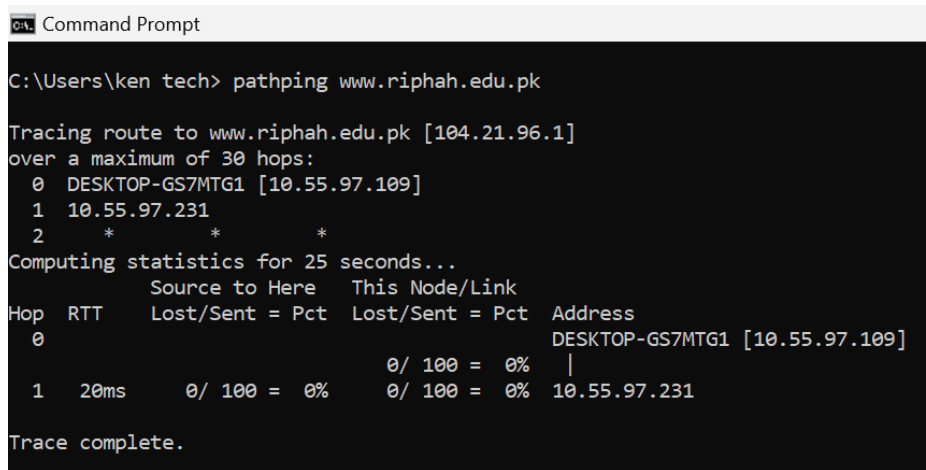
Trace complete.

C:\Users\ken tech>
```

Step 4: Last Practice

Also see “pathping ip or host” command. Which only shows path from source to destination.

The **PathPing** command is a command-line network utility supplied in Windows 2000 and beyond that combines the functionality of ping with that of tracert. It is used to locate spots that have network latency and network loss.



```
Command Prompt
C:\Users\ken tech> pathping www.riphah.edu.pk

Tracing route to www.riphah.edu.pk [104.21.96.1]
over a maximum of 30 hops:
  0  DESKTOP-GS7MTG1 [10.55.97.109]
  1  10.55.97.231
  2  * * *
Computing statistics for 25 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
  0                               0/ 100 = 0%      DESKTOP-GS7MTG1 [10.55.97.109]
    |
  1   20ms    0/ 100 = 0%      0/ 100 = 0%      10.55.97.231
Trace complete.
```

Close the window.

